

Alexey Khabalov

Head of AI Engineering / Director of AI & Data Platforms

FinTech | Applied AI | Cloud-Native Data Platforms | Engineering Leadership

EU passport holder

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Open to remote, hybrid, or relocation

PROFESSIONAL SUMMARY

Director-level engineering leader with 25+ years of experience building and scaling FinTech, cloud, data, AI-enabled, and distributed platforms across dxFeed, DataArt, KONE, and Philip Morris. Strong track record of turning executive strategy into production platforms, engineering organizations, technical roadmaps, and measurable business outcomes.

Most recently at dxFeed, directed data engineering and platform engineering for the company's first production-grade cloud-native Market Data Platform and AWS Data Lakehouse. Scaled the platform to 10PB+ of market and reference data, delivered 5 strategic initiatives within a \$1-2M scope, built and led a distributed team of 8 senior engineers, and expanded the platform into AI-enabled financial-data solutions through LLM, RAG, GraphRAG, and vector database PoCs.

For a Head of AI Engineering role, I bring a strong combination of AI platform direction, FinTech data foundations, cloud-native architecture, hands-on technical depth, engineering leadership, stakeholder management, governance, and production-delivery discipline. I am especially strong where AI must be connected to real business systems: financial data, customer workflows, backend platforms, observability, security, cost efficiency, and scalable engineering execution.

TARGET ROLE FIT

Head of AI Engineering

- Built production-grade FinTech data and analytics platforms from strategy to rollout.
- Led teams and platform roadmaps across cloud, data, backend, AI/ML PoCs, and customer-facing capabilities.
- Strong applied AI fit: RAG, GraphRAG, vector databases, LLM agents, AI-enabled financial-data workflows.
- Strong data-foundation fit: data lakehouse, ingestion, quality, APIs, entitlements, telemetry, billing, analytics.
- Strong organization-building fit: hiring, mentoring, engineering standards, roadmap execution, architecture governance, stakeholder alignment.
- Good fit for building AI engineering from zero to production impact, especially in a financial-services environment.

EXECUTIVE STRENGTHS

AI Engineering Leadership: Applied AI strategy, AI-enabled product delivery, RAG/GraphRAG systems, LLM agents, vector databases, AI platform PoCs, productionization planning

Data & Platform Strategy: Data Lakehouse, financial-data platforms, ingestion, data quality, data monetization, SQL/API access, entitlements, marketplace-style data products

Cloud-Native Architecture: AWS, Kafka, ClickHouse, Snowflake, Delta Lake, serverless patterns, distributed systems, microservices, APIs

Engineering Organization: Team building, hiring, mentoring, engineering standards, OKRs, roadmap execution, delivery governance, architecture reviews

Business & Stakeholders: Executive communication, board-level reporting, budgeting, P&L-aware; platform economics, cross-functional delivery, customer onboarding

Delivery & Operations: CI/CD, production readiness, incident prevention, observability, release governance, high-availability platforms

TECHNOLOGY HIGHLIGHTS

AI / LLM: LLM agents, RAG, GraphRAG, vector databases, Flowise, Neo4j, prompt-driven workflows, AI-enabled analytics

Data Engineering: Data Lakehouse, Spark, Delta Lake, Hudi prototypes, Airflow, SQL, data pipelines, ingestion adapters, data quality

Cloud: AWS S3, Athena, Glue, Lambda, SQS, SNS, EMR patterns, Azure Data Factory, GCP exposure

Backend & APIs: Java, Spring Boot, Node.js, TypeScript, Python, REST APIs, microservices, event-driven systems

FinTech: Market data, reference data, L1/L2 feeds, order books, trading analytics, entitlements, billing, telemetry

DevOps & Infra: Docker, Terraform, TeamCity, GitLab, Bitbucket, CI/CD, observability, release governance

PROFESSIONAL EXPERIENCE

dxFeed

dxFeed is a FinTech company providing real-time and historical market data, analytics, and trading infrastructure for brokers, exchanges, and institutional clients worldwide.

Senior Engineering Manager - Director-Level Scope

07/2022 - 12/2025 | Remote from Georgia / Europe

- Directed data engineering and platform engineering for the company's first production-grade cloud-native Market Data Platform, turning executive strategy into architecture, roadmap, OKRs, delivery model, and production rollout in under 12 months.
- Built and scaled an AWS Data Lakehouse with 10PB+ of market and reference data, enabling high-performance analytics, secure data access, AI/ML features, and productized data services.
- Delivered 5 strategic initiatives within an approximately \$1-2M budget scope, including secure SQL/API access, marketplace-style Data Shop services, ML Feature Store, Billing-Audit-Telemetry, and entitlement-enabled access.
- Built and led a distributed team of 8 senior engineers across Europe, improving delivery velocity and engineering quality through hiring, mentoring, architecture practices, and clear delivery governance.
- Expanded the platform into AI-enabled financial-data solutions through PoCs and early productionization work using LLMs, RAG, GraphRAG, vector databases, Flowise, and Neo4j.
- Designed platform capabilities for ingestion, validation, access control, customer onboarding, telemetry, and usage-based commercial operations.
- Reduced complex onboarding from weeks to hours by introducing standardized ingestion patterns, automation, configuration-driven workflows, and batch / near-real-time processing using Kafka and ClickHouse.
- Tracked platform economics and P&L-oriented; goals at company and department level while expanding monetization and usage-based service capabilities.
- Remained hands-on through architecture spikes and implementation work across TypeScript, Java, Python, Node.js APIs, Spark jobs, Airflow workflows, and Delta/Hudi prototypes.
- Partnered with executives, product stakeholders, engineering teams, and customer-facing functions to translate business needs into scalable platform and AI-enabled capabilities.

Leadership impact: Built a strategic FinTech data platform, led the engineering team, delivered commercial platform initiatives, and created a foundation for AI-enabled financial-data services.

DataArt

DataArt is a global software engineering and IT consulting company delivering digital, cloud, and data platforms for enterprise and regulated-industry clients.

Senior Technical Delivery Manager

09/2020 - 07/2022 | Russia / Armenia / Remote

- Directed modernization of legacy and greenfield data platforms for a leading U.S./global EdTech client across Azure Data Factory, AWS, GCP, Snowflake, and FiveTran.
- Managed 15 data engineers and QA on a legacy data platform while leading a new platform team of 10 engineers plus QA, BA, and Solution Architect.
- Managed ML-engineering and Data Ops workstreams in parallel, delivering several first production releases on time and within budget.
- Worked with customers, architects, engineering teams, and business stakeholders to define technical direction, remove delivery blockers, and ensure production adoption.
- Strengthened delivery governance, release planning, platform reliability, and cross-team coordination.

Projects Delivery Manager

01/2019 - 09/2020 | Russia / Remote

- Led a 9-month transformation project for a top Wall Street financial analytics client, migrating services from on-prem Red Hat/JBoss to cloud OpenShift/Kubernetes with a 20-person team.
- Managed multiple parallel projects for Swiss FinTech and startup clients across mobile, web, and digital-product domains.
- Delivered projects through distributed teams, stakeholder alignment, dependency management, and pragmatic architecture decisions.

Senior Release Manager

09/2017 - 12/2018 | USA, Argentina, Russia

- Owned production release orchestration for a top Wall Street financial analytics platform across 10+ teams and approximately 200 engineers.
- Managed up to 20 releases per week on lower environments and delivered 5 major releases in 15 months without rollback.
- Improved CI/CD, production readiness, release governance, communication, and acceptance processes with business stakeholders.

KONE International

KONE is a global engineering company delivering elevators, escalators, and smart-building solutions across urban infrastructure markets worldwide.

Product Owner / Expert

06/2016 - 12/2016 | Belgium

- Led New Model of Sales development, implementation, and rollout using React, MongoDB, and Android tablet devices for global field operations.
- Managed vendor delivery and product execution for a global sales platform used by internal teams and customer-facing workflows.
- Translated business requirements into roadmap, backlog, architecture priorities, and implementation plans.

Web Applications Manager / Expert

03/2007 - 06/2016 | Belgium

- Led enterprise platform leadership across kone.com, Global Intranet, Appian BPM, CAMOS, and PTC Servigistics-related changes.
- Managed internal and outsourced teams plus strategic vendors, supporting 45,000 users across 100 countries and 40 languages.
- Delivered programs within budgets up to EUR 2M/year and achieved at least 10% YoY operational savings through offshore and vendor optimization.
- Improved global system performance and rollout quality across complex enterprise infrastructure and distributed user bases.

Senior QA Analyst

06/2004 - 03/2007 | Belgium

- Led non-functional and performance testing for SAP Portal, SAP R/3 modules, and field-service mobile applications.
- Simulated global operations ahead of deployment waves across 40 countries, identified bottlenecks, and supported remediation.

Philip Morris International

Philip Morris International is a global consumer-products company operating large-scale finance, logistics, and commercial systems across multiple markets.

Senior Systems Analyst / Architect

03/2002 - 06/2004 | Russia

- Implemented and customized Oracle Portal for CIS commercial operations, integrating backend information systems with portal-based services.
- Supported Finance, Legal, Corporate Affairs, and commercial operations through enterprise-system delivery.

Systems Analyst

09/1999 - 03/2002 | Russia

- Delivered a Cash-Flow Management System for CIS Treasury, improving treasury operations and calculation flexibility.
- Built secondary logistics and factory reporting solutions, improving reporting quality and operational visibility.

IPP - Local Audit and Legal Company

IPP is a local Audit and Legal company where I worked across software development, architecture, project management, QA, and client delivery.

Software Developer / Architect / Project Manager

09/1996 - 09/1999 | Russia

- Designed and delivered two client-server systems for local utility and enterprise clients using Delphi, Java, MS SQL, and Crystal Reports.
- Worked across architecture, development, QA, business analysis, and deployment in a small team; both systems were delivered to production.

SELECTED AI / PLATFORM PROJECTS

AI-Enabled Financial Data Platform Work

- Led PoCs and early productionization paths for LLM-based agents connected to market-data workflows.
- Used RAG, GraphRAG, Flowise, Neo4j, and vector databases to explore AI-assisted retrieval, financial-data interpretation, and knowledge workflows.
- Connected AI experiments with real platform concerns: data access, entitlements, telemetry, cost, performance, observability, and production reliability.

AI Agent Pub Booking System - GitHub

<https://github.com/khab40/homework-pub-booking/>

- Designed and implemented a multi-step AI agent workflow for researching and booking a pub in Edinburgh using tool calls, structured validation, policy checks, and agent-to-dialog-system handoff.
- Demonstrates practical agent engineering patterns including tool orchestration, Rasa-based structured dialogue, atomic file IPC between loop and structured halves, dataflow-integrity checks, and optional voice interaction via

STT/TTS.

- Relevant to Head of AI Engineering roles because it shows applied AI system architecture beyond simple prompting: agent loops, structured validation, deterministic policy enforcement, observability through logs, and safe handoff between LLM and business-rule components.

LLM Architecture Lab - LoRA, RoPE, MoE - GitHub

<https://github.com/khab40/lm-arch-ht4>

- Implemented and documented modern LLM adaptation and architecture exercises, including LoRA from scratch for GPT-2, adapter save/load flows, PEFT comparison, RoPE attention, and sparse Mixture-of-Experts feed-forward layers.
- Demonstrates hands-on understanding of transformer internals, parameter-efficient fine-tuning, positional encoding, expert routing, PyTorch notebooks, Hugging Face tooling, and GPU-based experimentation.
- Relevant to Head of AI Engineering roles because it supports credibility in AI/LLM architecture discussions, model adaptation trade-offs, GPU experimentation, and communication with ML engineers.

Market Data Lake / Data Shop Demo - GitHub

<https://github.com/khab40/spring-boot-java-refresh>

- Designed and implemented a production-style Market Data Lake / Data Shop demo with Spring Boot, REST APIs, modular architecture, and extensible domain model.
- Demonstrates architecture for data ingestion, querying, service layering, API access, and entitlement-like patterns.
- Relevant to AI engineering leadership because it shows ability to build the backend/data foundation needed for AI-enabled financial products.

EDUCATION & CERTIFICATIONS

- MSc in Applied Mathematics and Physics - Peter the Great Polytechnic University
- SAFe Scrum Master
- ITIL v3
- Financial Markets - Yale University / Coursera
- Go
- Microservices with Node.js and React
- AI Agents with LangChain
- AI Performance Engineering at Nebius Academy in London - in progress

LANGUAGES

- English - Fluent professional
- Russian - Native
- French - Basic